Hospital Data Analysis – SQL Queries

Database: MySQL
Table: hospital_data

1. Total Patients

```
SELECT SUM(`Patients Count`) AS total_patients
FROM hospital_data;
```

2. Department-wise Patient Analysis

```
SELECT Department,
    SUM(`Patients Count`) AS total_patients
FROM hospital_data
GROUP BY Department
ORDER BY total_patients DESC;
```

3. Top 5 Hospitals by Patient Count

```
SELECT `Hospital Name`,
    SUM(`Patients Count`) AS total_patients
FROM hospital_data
GROUP BY `Hospital Name`
ORDER BY total_patients DESC
LIMIT 5;
```

4. Average Doctors per Hospital

```
SELECT `Hospital Name`,
    AVG(`Doctors Count`) AS avg_doctors
FROM hospital_data
GROUP BY `Hospital Name`
ORDER BY avg_doctors DESC;
```

5. Location-wise Patient Analysis

```
SELECT Location,
    SUM(`Patients Count`) AS total_patients
FROM hospital_data
GROUP BY Location
ORDER BY total_patients DESC;
```

6. Hospital with Highest Medical Expenses

```
SELECT `Hospital Name`,
    SUM(`Medical Expenses`) AS total_expenses
FROM hospital_data
GROUP BY `Hospital Name`
ORDER BY total_expenses DESC
LIMIT 1;
```

7. Average Length of Hospital Stay

```
SELECT AVG(DATEDIFF(`Discharge Date`, `Admission Date`))
    AS avg_stay_days
FROM hospital_data
WHERE `Admission Date` IS NOT NULL
 AND `Discharge Date` IS NOT NULL;
```

8. Monthly Medical Expenses

```
SELECT DATE_FORMAT(`Admission Date`, '%Y-%m') AS month,
    SUM(`Medical Expenses`) AS monthly_expenses
FROM hospital_data
WHERE `Admission Date` IS NOT NULL
GROUP BY DATE_FORMAT(`Admission Date`, '%Y-%m')
ORDER BY month;
```

9. Departments with More Than 1,000 Patients

```
SELECT Department,
    SUM(`Patients Count`) AS total_patients
FROM hospital_data
GROUP BY Department
HAVING SUM(`Patients Count`) > 1000
ORDER BY total_patients DESC;
```

10. Rank Hospitals by Patient Count (CTE + Window Function)

```
WITH hospital_summary AS (
  SELECT `Hospital Name`,
      SUM(`Patients Count`) AS total_patients
```

```
    FROM hospital_data
    GROUP BY `Hospital Name`
)
SELECT `Hospital Name`,
       total_patients,
       DENSE_RANK() OVER (
          ORDER BY total_patients DESC
       ) AS hospital_rank
FROM hospital_summary;
```

11. Department-wise Average Medical Expenses

```
SELECT Department,
       AVG(`Medical Expenses`) AS avg_medical_expenses
FROM hospital_data
GROUP BY Department
ORDER BY avg_medical_expenses DESC;
```

12. Hospital-wise Patient and Expense Summary

```
SELECT `Hospital Name`,
       SUM(`Patients Count`) AS total_patients,
       SUM(`Medical Expenses`) AS total_expenses,
       ROUND(
          SUM(`Medical Expenses`) /
          NULLIF(SUM(`Patients Count`), 0), 2
       ) AS expense_per_patient
FROM hospital_data
GROUP BY `Hospital Name`
ORDER BY total_patients DESC;
```